

DMFT and its connections to DFT

- ① core purpose.
 - ↳ DFT/DMFT strives to "solve" for the available electronic states of a system and "how" they are occupied. $\Rightarrow$ ground state energy / spatial distribution.
- ↳ if then questions: "temporal evolution" $\Rightarrow$ "perturbation".
 - ↳ how do these then affect the given system?
 - ↳ these are core questions of DMFT.

- ② DFT/DMFT
 - ① DFT/DMFT
 - ② code
 - ③ DE
 - ④ DMFT pipeline
 - ⑤ NAS (impurity NN).

③ DFT's "flow" (aka. SCF loop).

- 1) initial ncr guess = just averages each electron density.
- 2) calculate $\hat{H}(V_{\text{eff}})$ based on ncr.
- 3) plug in V_{eff} to KS eq. $[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{eff}}] \psi_i = \epsilon_i \psi_i$.
- 4) find ψ_i and ϵ_i .
- 5) use $\text{ncr} = \sum |\psi_i(r)|^2 \cdot f_i \rightarrow f_i$ shows how many electrons are in each ψ_i .
- 6) plug back in.

f_i = # of electrons assigned to ψ_i
 ψ_i = energetic assignment of ψ
 $\psi_i \psi_j$ = spatial assignment of electrons in ψ_i .

④ DFT's assumptions

- ↳ Schrödinger's cannot solve for a $3N$ -size system ($\exists (x,y,z) \times N$ electrons)
- ↳ assume condensed "electron density" which we make up from fictitious.

- full ex) ① set 2 kohn sham orbitals w/ 2 basis functions (χ_1, χ_2)
 so $\phi_1 = c_{11}\chi_1 + c_{21}\chi_2$ and $f_1=2, f_2=0$ (Pauli)
 $\phi_2 = c_{12}\chi_1 + c_{22}\chi_2$

② initial guess of ncr.

$$p^{(0)} = \begin{pmatrix} c_{11}^2 & c_{12}^2 \\ c_{21}^2 & c_{22}^2 \end{pmatrix} = \begin{pmatrix} 1.0 & 0 \\ 0 & 1.0 \end{pmatrix}$$

- Kohn-Sham orbitals $\{\psi_i\}$
- ↳ they stack up by eigenvalue index i .

③ build $\hat{H}_1 = -2.0 + 0.5(P_{12})$ (for each trial)

$$\hat{H} = \begin{pmatrix} -1.5 & -0.5 \\ -0.5 & -0.5 \end{pmatrix}$$

④ solve $\hat{H}c = \epsilon c$

$$\text{w/ } \hat{H} = \begin{pmatrix} -1.5 & -0.5 \\ -0.5 & -0.5 \end{pmatrix}$$

$$\hat{H} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \epsilon \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\hat{H}c_1 = \epsilon c_1$$

$$\hat{H}c_2 = \epsilon c_2$$

$$S = I$$

$$\hat{H}c_1 = \epsilon c_1$$

$$\psi_i = [-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{eff}}] \psi_i = \epsilon_i \psi_i$$
 where $\text{ncr} = \sum_{i=1}^N |\psi_i(r)|^2$ and $\text{ncr} \Rightarrow V_{\text{eff}} \Rightarrow \hat{H} \Rightarrow \epsilon_i$

what ncr minimizes $\hat{E}$?
 (ground state).
 b70s, s70s, etc... used to mimic spatial distribution after energetic distribution of eigen values.

- Kohn-Sham eq. solves for ψ_i and each ϵ_i
- ↳ also understood: $\psi_i = \sum c_j \chi_j$ where χ_j are basis functions.

- ψ_i (sorts on energy i)
- ↳ χ_j (sorts on spatial basis)

* derivation (comes from "minimization of energy")

- ① we take inner product $\langle \psi_i | \psi_j \rangle = \int \psi_i^* \psi_j$ for observables / orthogonality.

$$T_3 = -\frac{\hbar^2}{2m} \sum \int \psi_i^* \nabla^2 \psi_i d^3r$$
 and full energy
 Hamiltonian is:

$$E(\chi) = T_3 + E_{\text{ex}} + E_{\text{HF}} + E_{\text{xc}}$$

$$E_{\text{ex}} = \int \sum_{i,j} \frac{e^2}{4\pi\epsilon_0 |r-r'|} \text{ncr}(r) \text{ncr}(r') d^3r d^3r'$$

$$E_{\text{HF}} = \frac{e^2}{8\pi\epsilon_0} \iint \frac{\text{ncr}(r) \text{ncr}(r')}{|r-r'|} d^3r d^3r'$$

$$E_{\text{xc}} = \dots$$

E_{ex} : interaction w/ nucleus (attractive, $+/ -$)
 E_{HF} : electrostatic repulsion, self interaction (repulsive, $- / -$) (mean field approx)
 E_{xc} : others (localized interactions)

* gets Fermi levels $\Rightarrow$ can draw DOS.

⑤ we minimize this Lagrangian (under constraints)

- 1) minimize total energy
- 2) ensure $\int \psi_i^* \psi_j = \delta_{ij}$ (orthonormal)
- to satisfy multi-objective function we use a Lagrangian.
- * quick recap on the Lagrangian

$$\lambda = E - \lambda_6$$
 s.t when we minimize λ ($\frac{\partial \lambda}{\partial \epsilon} = 0$) it adjusts under constraint ϵ .

in this case,

$$\lambda = \int \psi_i^* \psi_j - \delta_{ij} = 0$$
 shows orthogonality + normalization.
 $\lambda = \epsilon_{ij}$ (two dimensional Lagrange multipliers)

acknowledge that ϵ_{ij} is an entangled matrix (not diagonalized).
 but ϵ_{ij} is Hermitian w/ we said $\epsilon_{ij} = \int \psi_i^* \psi_j$ and $\epsilon_{ji}^* = \int \psi_j^* \psi_i$
 s.t $\epsilon_{ji} = \epsilon_{ij}^*$ (complex conjugate)

Hermitian $\Leftrightarrow$ unitary so we apply a unitary transformation

ex) given ϕ_1 and ϕ_2
 create 2 new orbitals: $\phi_1' = \frac{\phi_1 + \phi_2}{\sqrt{2}}$, $\phi_2' = \frac{\phi_1 - \phi_2}{\sqrt{2}}$

$$\text{then new ncr} = |\phi_1'|^2 + |\phi_2'|^2$$

$$= \frac{\phi_1^2 + 2\phi_1\phi_2 + \phi_2^2}{2} + \frac{\phi_1^2 - 2\phi_1\phi_2 + \phi_2^2}{2}$$

$$= |\phi_1|^2 + |\phi_2|^2 = \text{ncr} \text{ so it's unitary.}$$

NOW, $\lambda = E - \sum \epsilon_i (\int \psi_i^* \psi_i - 1)$

$$\frac{\partial \lambda}{\partial \psi} = 0$$
 (minimize λ)

$$\frac{\partial \lambda}{\partial \psi} = \frac{\partial T_3}{\partial \psi} + \frac{\partial E_{\text{ex}}}{\partial \psi} + \frac{\partial E_{\text{HF}}}{\partial \psi} + \frac{\partial E_{\text{xc}}}{\partial \psi} - \frac{\partial}{\partial \psi} \left[\sum \epsilon_i \left(\int \psi_i^* \psi_i - 1 \right) \right] = 0$$

$$= -\frac{\hbar^2}{2m} \nabla^2 \psi_i + V_{\text{eff}} \psi_i + 0 - \epsilon_i \psi_i - 0 = 0$$

$$\therefore \left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{eff}} \right] \psi_i = \epsilon_i \psi_i \text{ DFT.}$$

④ DMFT's assumptions

→ locality (spatial dependence)

→ tries to solve for strongly correlated systems by still allowing time to be a varying factor.

→ DFT fails b/c it is mean field approximation.

It does not take into account "changing" environments. (DMFT only observes changes of one spatial grid.)

⑤ DMFT's basic knowledge.

1) Green's function

→ solves for a certain "impulse".

$\hat{G} = \int G(x, s) \delta(x-s)$ where $\delta(x-s)$ is the impulse at $x=s$
 $\hat{G}$ is the differential operator
 $\hat{G}$ is the "map"

$$\text{ex) } \frac{d^2}{dx^2} [G(x, s)] = \delta(x-s)$$

when $x \neq s$ ($x > s, x < s$) $\delta(x-s) = 0$.

$$G''(x, s) = 0.$$

$$\text{② assume } G(x, s) = \begin{cases} Ax+B & x > s \\ Cx+D & x < s. \end{cases}$$

③ apply spatial boundary conditions: ($0 < x < L$)
 $G(0, s) = 0, G(L, s) = 0$.

$$\therefore B = 0 \quad \therefore CL + D = 0, D = -CL$$

$$G(x, s) = \begin{cases} Ax \\ Cx - CL = C(x-L) \end{cases}$$

$$\text{④ } \lim_{x \rightarrow s^+} G(x, s) = \lim_{x \rightarrow s^-} G(x, s)$$

$$As = C(s-L)$$

⑤ to relate A to C. (small increment out of s)

$$\int_{s-\epsilon}^{s+\epsilon} G''(x, s) dx = \int_{s-\epsilon}^{s+\epsilon} \delta(x-s) dx$$

$$[G'(x, s)]_{s-\epsilon}^{s+\epsilon} = -1$$

$$G'(x, s) = \begin{cases} A & x < s \\ C & x > s \end{cases}$$

$$\text{so } C - A = -1$$

⑥ combine $C - A = -1$ and $As = C(s-L)$

$$A = C + 1, Cs + s = C(s-L)$$

$$s = -CL, C = -\frac{s}{L}, A = \frac{L-s}{L}, D = -CL = \frac{s}{L} \cdot L = s$$

$$G(x, s) = \begin{cases} \frac{L-s}{L} x \\ -\frac{s}{L} x + s. \end{cases}$$

$$\text{⑦ } u(x) = \int_0^L G(x, s) f(s) ds$$

total response. $\rightarrow$ force at x
 $\rightarrow$ all point responses.

2) form of Green's Function.

deriving $\hat{G} = \omega - \hat{H}$ (where ω will not be frequency)

① TDSE

$$i \frac{\partial}{\partial t} \psi = H \psi$$

$$\left(i \frac{\partial}{\partial t} - H \right) \psi = 0.$$

$$\hat{G} \psi = 0.$$

② set up for Green's function

$$\hat{G} G(x, t) = \delta(x-t)$$

③ Fourier transform $G(x, t)$ and $\delta(x-t)$

$$\Rightarrow G(x, t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega(t-t')} G(\omega)$$

$$\Rightarrow \delta(x-t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega(t-t')} I$$

$$\begin{aligned} \left(i \frac{\partial}{\partial t} - H \right) \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega(t-t')} G(\omega) &= \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega(t-t')} I \\ \text{LHS} &= \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \left[i \frac{\partial}{\partial t} e^{-i\omega(t-t')} G(\omega) - H e^{-i\omega(t-t')} G(\omega) \right] \\ &\Rightarrow -i\omega e^{-i\omega(t-t')} G(\omega) \cdot i \Rightarrow e^{-i\omega(t-t')} \hat{H} G(\omega) \\ &\Rightarrow \omega e^{-i\omega(t-t')} G(\omega) \\ &= \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega(t-t')} [\omega - \hat{H}] G(\omega) \end{aligned}$$

④ combine w/ RHS

$$\int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega(t-t')} [\omega - \hat{H}] G(\omega) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega(t-t')} I$$

$$\text{so } (\omega - \hat{H}) G(\omega) = I$$

$$\text{s.t. } [\omega - \hat{H}]^{-1} = G(\omega)$$

$$G(\omega) = \frac{1}{\omega - \hat{H}} \therefore$$

3) DOS projection from Green's function.

$$\Rightarrow G(\omega) = (\omega - \hat{H} + i\eta)^{-1}$$

where $i\eta$ prevents denominator $\rightarrow 0$
 let $\omega - \hat{H} = x$.

$$\frac{1}{x+i\eta} \cdot \frac{(x-i\eta)}{(x-i\eta)} = \frac{x-i\eta}{x^2+\eta^2}$$

$$= \frac{x}{x^2+\eta^2} - i \left(\frac{\eta}{x^2+\eta^2} \right)$$

$$\text{Im} \left(\frac{1}{x+i\eta} \right) = \frac{\eta}{x^2+\eta^2}$$

$$\int_{-\infty}^{\infty} \frac{\eta}{x^2+\eta^2} dx \Rightarrow \text{spikes when } x=0, \eta \rightarrow 0^+$$

$$= \int_{-\infty}^{\infty} \text{arctan} \left(\frac{x}{\eta} \right) dx = \pi$$

$$= \pi \int_{-\infty}^{\infty} f(x) dx \text{ s.t. } f = \text{derivative}$$

$$= \pi \int G(\omega - \hat{H})$$

$$\text{Tr } G(\omega) = \sum \frac{1}{\omega - E_n + i\eta}$$

$$\Rightarrow \sum -\pi \delta(\omega - E_n)$$

$$\text{DOS} = \sum \delta(\omega - E_n)$$

$$\frac{-1}{\pi} \cdot \text{Im Tr } G(\omega) = \text{DOS}$$

exponential reality.
 (to be able to observe it)

⑥ self-energy / Dyson's Eq.

$\Sigma(k, \omega)$ = self-energy (all energy acting on same particle).

↳ In a non-interacting system, we

$$\text{define } G_0(k, \omega) = \frac{1}{\omega - E_n + i\eta}$$

↳ In an interacting system, electrons repel other electrons, leading to a quasi-particle.

$$\Sigma(k, \omega) = \text{scattering shift} + \text{scattering}$$

$$\boxed{\Sigma = G_0^{-1} - G^{-1}} \text{ equation, where } G_0 = \frac{1}{\omega - E_n + i\eta}$$

↳ technically, $G^{-1} = G_0^{-1} - \Sigma$

$$G = (G_0^{-1} - \Sigma)^{-1} = \frac{1}{G_0^{-1} - \Sigma} \quad \text{since } -\frac{1}{\omega} \text{Im } G = \text{DOS}$$

It can be used to find DOS.

⑦ DMFT "flow".

1) get G_{imp} first. solves for a single propagation response that is non-interacting.

2) Matsubara frequency is set before initial $\Sigma^{(0)}$ guess.

$$\omega_n = (2n+1)\pi \quad (\text{these are the odd points})$$

* compare quantum mech / statistical mech.

$$\text{quantum mech} = e^{-\frac{\hat{H}\tau}{\hbar}}$$

$$\text{stat mech} = e^{-\beta H} \quad (\text{probability at temp } \beta \text{ and energy } E).$$

$$\rightarrow Z = \text{Tr}(e^{-\beta H}) \quad (\text{sum over all microstates})$$

↳ we define: $t = -i\hbar\tau$

$$U(t) = e^{-\frac{i\hbar^{-1} \hat{H} t}{\hbar}} = e^{-\hat{H} t}$$

If we set $0 < t < \beta$.

then a quantum system evolving through $t = \beta$ is same as a thermal system at $T = \beta$.

↳ now, if $\beta = \frac{1}{k_B T}$ as $T \uparrow \beta \downarrow$ time loop = short
 $T \downarrow \beta \uparrow$ time loop = long.

$$\rightarrow \omega_n = \frac{2n\pi}{\beta} \quad \text{bosons / fermions.}$$

3) for example since $\omega_n = \frac{(2n+1)\pi}{\beta}$ $\rightarrow$ thermal energy quantum available $\omega_0 = 0.34$

$$\Sigma^{(0)}(0.314) = 0.0 + 0.0 \quad \text{basically projecting } \Sigma^{(0)} \text{ to rest of lattice}$$

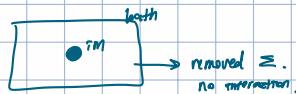
$$G_{\text{loc}}(0.314) = \int_{-\infty}^{\infty} \frac{P(E)}{0.314 + i\eta - E - \Sigma(0.314)} dE \quad \text{to see how they propagate.}$$

"local lattice's total energy map"

4) Weiss cavity (isolated "bath")

↳ calculate G_0 (non-interacting part).

$$G_0(\tau) = \frac{1}{G_{\text{loc}}^{-1}(\tau) + \Sigma(\tau)}$$



What would happen when nothing is injected initially?

Hybridization function

$$\Delta(\tau) = \tau \omega_n + \mu - E_d - G_0^{-1}(\tau)$$

5) impurity solver. (ex, $\alpha\text{-NAC}$). What is interacting energy on that SINGLE site. with U turned on

$$G_{\text{imp}}(\tau) = \frac{1}{\beta} \sum_{\omega_n} e^{-i\omega_n \tau} G_{\text{imp}}(\omega_n)$$

$$H_{\text{A2M}} = H_{\text{loc}} + H_{\text{bath}} + H_{\text{hyb.}} \quad \text{impurity + bath states.}$$

d^+ : leaves bath, lands on grid (impurity)

d : leaves grid, back into bath.

since we have μ chemical potential = likelihood for electron to leave center.

$$\text{If } G_{\text{isolated}} = \tau \omega_n + \mu - E_d \rightarrow \text{you calculate "relative energy loop"}$$

$$\Delta(\tau) = \sum_k V_k \left(\frac{1}{\tau \omega_n + \mu - E_d} \right) V_k$$

hopping on and off

6 rec's (propagator)

M^{-1} = $K \times K$ matrix of all interactions.

$$M_{ij}^{-1} = \begin{pmatrix} \Delta(\tau_1 - \tau'_1) \\ \Delta(\tau_2 - \tau'_2) \\ \vdots \end{pmatrix}$$

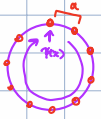
Fourier transformed to imaginary time

⑧ Quantum Espresso

↳ Based on periodic system

↳ uses Bloch states as basis set.

↳ Bloch's Theorem



N dots

$$\psi(x) = \psi(x + Na)$$

$$\psi(x + Na) = e^{iNka} \psi(x)$$

$$e^{iNka} = 1 \rightarrow e^{iNka} = e^{i2\pi s}, s \in \mathbb{Z}$$

constrained to N points

$$e^{iNka} = e^{i2\pi s}$$

$$ka = \frac{2\pi s}{N}, k = \frac{2\pi s}{Na} \text{ so } k \text{ travels by } \frac{2\pi}{Na}$$

this periodicity condition is the same as a plane wave restricted by a translation of $\frac{2\pi}{Na}$.

all hopping on and offs at state k .
enters at i , exits at j

$$W = \det M^{-1} \times \text{Tr}_{\text{occ}} [e^{-\beta H}] \text{ (dd*)}$$

$$\frac{e^{-\beta E}}{\text{partition factor}}$$

↳ score

↳ accept/reject hop

$$\rightarrow \min(1, \frac{W_{\text{new}}}{W_{\text{old}}})$$

* M.C theory #.

Wannier Functions (DFT + DMFT)

↳ Bloch states $\leftrightarrow$ Wannier functions

↳ Recall that Bloch states are used as a basis set in the SCF loop when we are building the Kohn-Sham orbitals:

$$\Phi = \sum_i c_i \chi_i$$

↳ thus, we are constructing a Fourier Transform between k -space and real space.

$$W(r) = \frac{V}{(2\pi)^3} \int_{BZ} d^3k \cdot e^{-i\mathbf{k} \cdot \mathbf{r}} \psi_{\mathbf{k}}(r)$$

↳ must be orthonormal

↳ centered on atom

↳ must decay in real space.

↳ thus, the phase assigned cannot be random. (MLWFs)

$$\textcircled{1} \rightarrow \sum_N \langle P \rangle_N - \underbrace{1}_{\text{expectation value at position}} \underbrace{f_{\mathbf{k}}}_{\text{com.}}$$

→ how for FS com. from the overall prop

approx.

↳ What is "gauge"?

↳ the set of "phases" for the BZ calculation.
ex) when you have.

$$u_{\mathbf{k}} = e^{i\mathbf{k} \cdot \mathbf{r}} \cdot \psi(\mathbf{k})$$

the phase factor $e^{i\mathbf{k} \cdot \mathbf{r}}$ can be restricted.

↳ Wannier functions are gauge-dependent b/c reciprocal space defines all $\psi(\mathbf{k})$ translated by $e^{i\mathbf{k} \cdot \mathbf{r}}$ to be identical

since $V_{BZ} = \frac{(2\pi)^3}{V_{cell}}$

this is b/c.

$$V_{cell} = a_1 \cdot (a_2 \times a_3)$$

and since $a_i \cdot b_j = 2\pi \delta_{ij}$
(reciprocal definition)

let $A = \begin{pmatrix} a_{1x} & a_{1y} & a_{1z} \\ a_{2x} & a_{2y} & a_{2z} \\ a_{3x} & a_{3y} & a_{3z} \end{pmatrix}$ and $B = \begin{pmatrix} \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } \end{pmatrix}$

$$\det(A) = V_{cell} \quad \det(B) = V_{BZ}$$

since $\det(AB^T) = a_i \cdot b_j = 2\pi I_3$

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} (b_1 \ b_2 \ b_3) = \begin{pmatrix} 2\pi & & \\ & 2\pi & \\ & & 2\pi \end{pmatrix}$$

$$\begin{aligned} \det(AB^T) &= \det(A) \cdot \det(B^T) \\ &= \det(A) \cdot \det(B) = 2\pi I_3 \\ &= V_{cell} \cdot V_{BZ} = 2\pi I_3 \end{aligned}$$

$$\therefore V_{BZ} = \frac{2\pi I_3}{V_{cell}} = \frac{(2\pi)^3}{V_{cell}}$$

but not ψ in real space.

so how do we find the appropriate gauge? $\rightarrow$ here, the gauge is:

$$U_k(r) = U \cdot \psi_k(r)$$

where

$$w(r) = \sum e^{-ikr} \left[\underbrace{U \cdot \psi_k(r)}_{\text{we need gauge } U \text{ to be smooth}} \right]$$

(2)

$$p = -i\hbar \frac{\partial}{\partial x}$$

$$p = \hbar k$$

since

$$e^z = 1 + z + \frac{z^2}{2!} + \dots = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

it means

$$\psi(z-\varepsilon) = \left[1 - \varepsilon \frac{d}{dx} + \frac{\varepsilon^2}{2!} \frac{d^2}{dx^2} \dots \right] \psi(x)$$

$$= e^{-\varepsilon \frac{d}{dx}} \psi(x)$$

$$\text{therefore } = T(\varepsilon) \psi(x)$$

$$T(\varepsilon) = e^{-\varepsilon \frac{d}{dx}}$$

$$1 - \frac{i}{\hbar} d\varepsilon p = \left[-d\varepsilon \frac{d}{dx} \right]$$

$$-\frac{i}{\hbar} p = \frac{d}{dx}$$

$$\therefore p = -i\hbar \frac{d}{dx}$$

$\rightarrow$ since $p = -i\hbar \nabla_r$
 $r = -i\hbar \nabla_k$.

so to calculate r for Ω we need ∇_k .

$$U_{k+b}(r) \approx U_k(r) +$$

shifting by b .

$$(b \cdot \nabla_k) U_k(r)$$

same thing as shifting by $b \cdot \nabla_k$

$$M_{k, k+b} = \langle U_k | U_{k+b} \rangle = \langle U_k | U_k \rangle + b \cdot \langle U_k | \nabla_k U_k \rangle$$

$$= 1 + b \cdot \langle U_k | \nabla_k U_k \rangle$$

$$= 1 + b \cdot -i\hbar \bar{k}$$

$$= 1 - i\hbar (F_r \cdot b)$$

$$= e^{-i\hbar (F_r \cdot b)}$$

$$\text{Im}(\ln M_{k, k+b}) = -b \cdot \bar{k}$$

(3) $\Omega = \Omega_I + \Omega_D$ we need $\Omega_D \rightarrow 0$

this is how we get smooth U matrix.
but we must realize that this requires trial states.

$$v_k = \frac{u_{k+b} - u_k}{b}$$

$$M = \langle u_k | u_{k+b} \rangle \neq \text{overlap matrix}$$

$$|M| = \Omega \rightarrow \text{update } \Omega \rightarrow \text{find } U \rightarrow F7$$

- 1) i need to find wannier functions (basis sets) in real space instead of k space
- 2) simple FT will distort it because not all phases will produce smooth functions. so we need a gauge set that picks only specific phases.
- 3) To do this we need to find a matrix U that holds all of these gauges.
- 4) To do that we learn that we must minimize the spatial variance in order to focus the phases/ electronic wavefunctions in a very dense grid.
- 5) to get spatial variance we need del operator k
- 6) to get del operator k we use $u_{k+b} - u_k / b$ and then use M as a n inner product between them to get spatial variance.
- 7) once spatial variance is minimized we see if it is the actual minimal. so the U that we selected for that calculation will be chosen if the spatial variance was minimized

Neural Quantum States (NQS) for Impurity Solvers.

1) What are neural quantum states (NQS)?

→ Quantum state $|\psi\rangle$ gives information on the probability amplitude of finding that state in a specific position and spin $\uparrow, \downarrow$.

→ consider $\psi(r, b)$ where $\begin{cases} r = \text{spatial pos.} \\ b = \text{spin} \end{cases}$

→ We separate $\psi(r, b)$ into M discrete spatial orbitals such that

$$\{\phi_1(r, b), \phi_2(r, b), \dots, \phi_M(r, b)\}$$

→ Fock space is an occupancy list for each orbital:

$$|x\rangle = | \phi_1(r, b) |^2, | \phi_2(r, b) |^2, \dots, | \phi_M(r, b) |^2$$

ex) $\psi(r, b) \rightarrow \begin{cases} \phi_1(r, b) A, \uparrow \\ \phi_2(r, b) A, \downarrow \\ \phi_3(r, b) B, \uparrow \\ \phi_4(r, b) B, \downarrow \end{cases}$ r^2 arrangements.

→ Inject 1 electron $\psi(r, b)$ into a system of 4 orbitals.

probability $p_1 + p_2 + p_3 + p_4 = 1$
 $p_1 = p_2 = p_3 = p_4, p_i = \frac{1}{4}$
 since $p = |\psi|^2, c = \sqrt{\frac{1}{4}} = \frac{1}{2}$

$$\text{so } \psi(r, b) = \frac{1}{2} |1, 0, 0, 0\rangle + \frac{1}{2} |0, 1, 0, 0\rangle + \frac{1}{2} |0, 0, 1, 0\rangle + \frac{1}{2} |0, 0, 0, 1\rangle$$

→ Inject 2 electrons
 $\begin{pmatrix} \text{orbital \#} \\ \text{electron \#} \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} = 6$
 $\sum p = 6, p = \frac{1}{6}$
 $c = \sqrt{\frac{1}{6}}$

→ non-Fock space representation:

Slater determinant: $\psi(x_2, x_1) = -\psi(x_1, x_2)$

if $\psi(x_1, x_2) = \frac{1}{\sqrt{2}} \det \begin{pmatrix} \phi_1 & \phi_2 \\ \phi_3 & \phi_4 \end{pmatrix}$
 $= \frac{1}{\sqrt{2}} (\phi_1 \phi_4 - \phi_2 \phi_3)$

$\psi(x_2, x_1) = \frac{1}{\sqrt{2}} \det \begin{pmatrix} \phi_3 & \phi_4 \\ \phi_1 & \phi_2 \end{pmatrix}$
 $= \frac{1}{\sqrt{2}} (\phi_3 \phi_2 - \phi_4 \phi_1)$

so $H = \begin{bmatrix} & 210 \\ 210 & \end{bmatrix}$

so you need to make a determinant for every interaction.

(electron #) x (electron #) matrix

Items and

$$H_{1,2} = \int d^3r_1 d^3r_2 d^3r_3 d^3r_4 [\det(S_1)^* H \det(S_2)]$$

each entry of going from config 1 to config 2.

there are $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ configurations.

like $\begin{pmatrix} 4 \\ 2 \end{pmatrix} = 6$ $\begin{matrix} \phi_1 & \phi_2 & \phi_3 & \phi_4 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{matrix}$

$$\begin{array}{l} |0,0,1,1\rangle \\ |1,0,1,0\rangle \\ |1,0,0,1\rangle \\ |0,1,1,0\rangle \\ |0,1,0,1\rangle \end{array} \quad \text{configurations.}$$

Thus for Hamiltonian I would need configuration $\times$ configuration matrix.

for Fock

$$H = \sum h_{ij} c_i^\dagger c_j + \frac{1}{2} \sum V_{ijkl} c_i^\dagger c_j^\dagger c_l c_k$$

electron from orbital $i \rightarrow j$
 $\hookrightarrow$ 1-body (just electron moving)

Before: kl (annihilated)
 after: ij (created)

$\hookrightarrow$ 2-body: electron-electron repulsion

$\left. \begin{array}{l} e_1: k \\ e_1: i \end{array} \right\} \Leftrightarrow \left. \begin{array}{l} e_2: l \\ e_2: j \end{array} \right\} \text{ moved due to repulsion.}$

$$h_{ij} = \int d^3r \phi_i^*(r) H \phi_j(r)$$

$$V_{ijkl} = \iint d^3r_1 d^3r_2 \phi_i^*(r_1) \phi_j^*(r_2) \frac{1}{|r_1 - r_2|} \phi_l(r_1) \phi_k(r_2)$$

$\hookrightarrow$ Solve Lanczos.

$Hc = E_c$ solve for c .

c is then used as coefficients for Fock space.

applying Fock space to Quantum state representation:

$$|\psi\rangle = \sum_x c_x |x\rangle$$

$$\downarrow$$

$$\sum_x \psi_\theta(x) |x\rangle$$

this happens by $z_1 = w_1 b_1 + b_1$

try guess $\rightarrow c_x = \psi_\theta(x) = e^{\frac{\ln \psi_\theta(x) + \gamma \phi_\theta(x)}{\text{tries to scale down}}}$

phase changes which are the spatial derivative of electron flow.

$$E(\theta) = \frac{\langle \psi_\theta | H | \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle} \quad \text{high dimensional representation}$$

Must minimize $E(\theta)$.

Loss mainly depends on this variational principle.

$$\begin{aligned} H_{\text{loc}} &= \frac{\langle x | H | \psi_\theta \rangle}{\langle x | \psi_\theta \rangle} \\ &= \frac{\langle x | H | \psi_\theta(x') \rangle}{\langle x | \psi_\theta(x') \rangle} \end{aligned}$$

$$= \frac{\sum \langle x | A | x' \rangle \cdot \psi_\theta(x)}{\langle x | x' \rangle \psi_\theta(x')}$$

$$= \sum A_{x,x'} \frac{\psi_\theta(x)}{\psi_\theta(x')}$$

ex).

① $|\psi\rangle = \underbrace{\psi_\theta(x)}_{\text{coeff of each}} \underbrace{|x\rangle}_{\text{basis set.}}$ usually $\phi_1, \phi_2, \phi_3, \phi_4$

$$|x\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

② $\psi_\theta(x) = \exp(w \cdot D(x) + i\phi C(x))$ and $c_1, 1, 0, 0, 0$

$w \cdot D(x) = \ln |\psi_\theta(x)|$
 $\phi \cdot C(x) = \phi_\theta(x)$

$\psi_\theta(x) = \exp(\ln |\psi_\theta(x)| + i\phi_\theta(x))$ makes up $|\psi\rangle$.

$c_2, 1, 0, 1, 0, 0$
 $c_3, 1, 0, 0, 1, 0$
 $c_4, 1, 0, 0, 0, 1$

relates to electron flow

③ set up H

$$H = \begin{pmatrix} |x_1\rangle & |x_2\rangle & |x_3\rangle & |x_4\rangle \\ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} & \begin{pmatrix} 2.0 & 0.0 \\ 0.0 & 2.0 \end{pmatrix} & \dots & \dots \end{pmatrix}$$

going from one state to the other.

④ connecting the 2 concepts

now I have (w, ϕ) as my "params".

$D(x)$ = double occupancy sensor

$C(x)$ = single occupancy sensor.

$w \rightarrow$ how does $D(x)$ affect $\ln |\psi_\theta(x)|$

$\phi \rightarrow$ how does $C(x)$ affect $i\phi_\theta$

ex) $w = 0.0, \phi = 0.0$

$$\psi(x_1) = \exp(0.0 \cdot 1 + i \cdot 0.0 \cdot 0) = 1 + 0 \cdot i$$

$$p(x_1) = \frac{1}{4} = \frac{|\psi(x_1)|^2}{\sum |\psi|^2}$$

probability of particular at x / total possibilities.

$$H_{loc} = \sum_{x,x'} \frac{\psi(x')}{\psi(x)}$$

$$H_{loc}(x_1) = H_{1,1} \frac{\psi(x_1)}{\psi(x_1)} + H_{1,2} \frac{\psi(x_2)}{\psi(x_1)} + H_{1,3} \frac{\psi(x_3)}{\psi(x_1)}$$

$$E = \sum p(x_i) H_{loc}(x_i) \rightarrow \text{average weighting.}$$

Back prop. $\frac{\partial \text{loss}}{\partial \text{weight.}} = \frac{\partial \ln \psi(x)}{\partial w}$ or $\frac{\partial \ln \psi(x)}{\partial \phi}$

$$= 0$$

$$\partial r = \tau C$$

$$V = 2 \sum p_{\omega} \times p_{\omega} [A_{\omega} - E \times O_{\omega}^*]$$

ex)

$$2 \times 0.25 \times p_{\omega} [(1-\tau) \times (-\tau)]$$

$$= 2 \times 0.25 \times 1.0 = 0.5$$

$$W_{new} = w_{old} - \eta V \quad \text{S.t.} \quad W_{new} = 0 - (0.1)(0.5) \\ = -0.05$$